

Direction A: Richer Process Models for Kalman Associative Memory

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1 Question and scope

Kalman associative memory separates the evolution of memory from the correction made by a new key–value observation. Its linear–Gaussian model is

$$\mathbf{S}_t^* = \mathbf{D}_t \mathbf{S}_{t-1}^* + \mathbf{W}_t, \quad \mathbf{W}_t \sim \mathcal{N}(0, \boldsymbol{\Omega}_t), \quad (1)$$

$$\mathbf{v}_t = \mathbf{S}_t^{*\top} \mathbf{k}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(0, r_t \mathbf{I}). \quad (2)$$

The Kalman prediction is

$$\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}, \quad \hat{\mathbf{P}}_t = \mathbf{D}_t \mathbf{P}_{t-1} \mathbf{D}_t^\top + \boldsymbol{\Omega}_t. \quad (3)$$

The usual discussion interprets $\mathbf{D}_t$ as a forgetting operator. More precisely, vanilla DeltaNet has $\mathbf{D}_t = \mathbf{I}$ and does not impose global geometric forgetting; later delta writes can still damage an old binding when their keys overlap it. Explicit geometric decay enters Gated DeltaNet through $\mathbf{D}_t = \alpha_t \mathbf{I}$ and KDA through $\mathbf{D}_t = \text{diag}(\boldsymbol{\alpha}_t)$.

Diagnostics from the current GDN-2 model make the question concrete. Its median learned amplitude half-life is seven tokens, while only 16.9% of channels exceed 512 tokens and 7.8% exceed 4096 tokens. At length 8192, the mean effective rank of the memory is 18.1 out of 128. These measurements motivate a persistent subspace, but do not by themselves prove that decay is the only cause: low-dimensional key and write geometry can also produce low state rank. The hypothesis here is therefore not merely “add slower channels.” It is to give surviving information a controllable latent geometry and route selected content through it.

The Kalman view permits a broader question. Instead of only asking how quickly a stored association should shrink, can the process model move it into a latent subspace that has little effect on current predictions, preserve it there, and make it visible to a later query? A useful richer process must control three properties separately:

Property	Controlled by	Meaning
survival	transition-product norm	how long state amplitude remains
phase	rotation angle	how state moves among latent coordinates
visibility	write/read routing	whether a query currently observes it

For the normal rotation blocks developed below, the transition-product norm is exactly the product of their radii. Spectral radius alone would not control finite-time survival for a general non-normal or time-varying transition. Positive diagonal decay already permits visibility changes through a changing query; what it lacks is a process mechanism that moves surviving information between latent directions. This note develops a structured linear $\mathbf{D}_t$ with that mechanism. A nonlinear

state-dependent process $f_t(\mathbf{S}_{t-1})$ is possible in an extended-filter formulation, but it gives up the exact Kalman recursion and most of the current scan structure. The first step should therefore remain linear in the recurrent state, while allowing its coefficients to be token dependent and known before the memory scan.

2 Geometric decay cannot hide by state motion

Consider the contribution of one innovation written at time τ ,

$$\mathbf{U}_\tau = \boldsymbol{\kappa}_\tau \boldsymbol{\delta}_\tau^\top, \quad \boldsymbol{\delta}_\tau = \mathbf{v}_\tau - \widehat{\mathbf{S}}_\tau^\top \mathbf{k}_\tau. \quad (4)$$

To isolate the process model, first suppress all intervening measurement corrections. The hypothetical process-only contribution at time $t > \tau$ is

$$\widetilde{\mathbf{S}}_{t \leftarrow \tau}^{\text{proc}} = \boldsymbol{\Phi}_{t:\tau} \boldsymbol{\kappa}_\tau \boldsymbol{\delta}_\tau^\top, \quad \boldsymbol{\Phi}_{t:\tau} = \mathbf{D}_t \mathbf{D}_{t-1} \cdots \mathbf{D}_{\tau+1}. \quad (5)$$

Applying a query to this process-only state gives

$$\widetilde{\boldsymbol{o}}_{t \leftarrow \tau}^{\text{proc}} = \boldsymbol{\delta}_\tau \underbrace{\boldsymbol{\kappa}_\tau^\top \boldsymbol{\Phi}_{t:\tau}^\top \mathbf{q}_t}_{\text{visibility of write } \tau \text{ at time } t}. \quad (6)$$

Equation (6) distinguishes process survival from process-only visibility. The norm $\|\boldsymbol{\Phi}_{t:\tau} \boldsymbol{\kappa}_\tau\|$ measures how much of the write survives, whereas its alignment with $\mathbf{q}_t$ measures how much is currently observable.

For KDA,

$$\boldsymbol{\Phi}_{t:\tau} = \text{diag} \left(\prod_{s=\tau+1}^t \alpha_{s,1}, \dots, \prod_{s=\tau+1}^t \alpha_{s,d_k} \right). \quad (7)$$

When $0 \leq \alpha_{s,i} \leq 1$, the magnitude of every scalar mode is nonincreasing. A fast mode becomes exponentially small and cannot be restored by the process alone. A slow mode survives, but it is not hidden by the process: it remains in the same coordinate and is visible whenever the query overlaps that coordinate. Different α_i therefore form a bank of decay timescales, not a bank of Fourier frequencies. A fixed basis rotation $\mathbf{U} \text{diag}(\boldsymbol{\alpha}) \mathbf{U}^\top$ still has only positive real eigenvalues and the same zero-frequency exponential modes in a different coordinate system.

The exact posterior-mean recurrence includes every intervening residual correction. Expanding the Kalman update gives

$$\mathbf{S}_t = \underbrace{(\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{k}_t^\top) \mathbf{D}_t}_{\mathbf{A}_t} \mathbf{S}_{t-1} + \boldsymbol{\kappa}_t \mathbf{v}_t^\top. \quad (8)$$

Define the closed-loop propagation operator

$$\boldsymbol{\Psi}_{t:\tau} = \mathbf{A}_t \mathbf{A}_{t-1} \cdots \mathbf{A}_{\tau+1}. \quad (9)$$

The actual contribution of write τ to the posterior output at time t is therefore

$$\mathbf{o}_{t \leftarrow \tau} = \boldsymbol{\delta}_\tau \boldsymbol{\kappa}_\tau^\top \boldsymbol{\Psi}_{t:\tau}^\top \mathbf{q}_t. \quad (10)$$

The process-only response in (5) isolates what the choice of $\mathbf{D}_t$ contributes; (10) is the retrieval quantity the full layer must preserve. For hidden modes to survive, ordinary measurement keys must weakly observe those modes, or later residual corrections will erase and mix them.

3 A damped rotational process model

General tunable phase requires two-dimensional real planes; a scalar negative pole is the special Nyquist case $\theta = \pi$. Define the real rotation

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (11)$$

Partition the memory rows into M fixed mode groups. Mode m contains two quadratures of an r_m -dimensional content address and uses

$$\mathbf{D}_t^{(m)} = \rho_{t,m} \mathbf{R}(\theta_{t,m}) \otimes \mathbf{I}_{r_m}, \quad 0 < \rho_{t,m} \leq 1, \quad (12)$$

with

$$\mathbf{D}_t = \text{blockdiag}(\mathbf{D}_t^{(1)}, \dots, \mathbf{D}_t^{(M)}). \quad (13)$$

An optional fixed orthogonal basis $\mathbf{U}$ gives $\mathbf{U} \mathbf{D}_t \mathbf{U}^\top$; in a neural layer this basis can usually be absorbed into the key and query projections rather than applied to the recurrent state at every token.

Because all rotations within one fixed plane commute, the transition product remains simple:

$$\Phi_{t:\tau}^{(m)} = \left(\prod_{s=\tau+1}^t \rho_{s,m} \right) \mathbf{R} \left(\sum_{s=\tau+1}^t \theta_{s,m} \right) \otimes \mathbf{I}_{r_m}. \quad (14)$$

For constant ρ_m and θ_m , the eigenvalues are the complex-conjugate pair $\rho_m e^{\pm i\theta_m}$ in the closed unit disk. The homogeneous dynamics are asymptotically stable for $\rho_m < 1$ and only marginally stable for $\rho_m = 1$. The radius ρ_m sets lifetime. For $\rho_m < 1$, the amplitude half-life is

$$L_{1/2,m} = \frac{\log 2}{-\log \rho_m}, \quad (15)$$

while $\rho_m = 1$ gives an infinite half-life. The angle θ_m sets frequency and phase evolution. These are independent parameters.

For a scalar content direction, let $\mathbf{b}_m \in \mathbb{R}^2$ be its write phase and $\mathbf{c}_m \in \mathbb{R}^2$ its read phase. The process impulse response is

$$h_m(\ell) = \mathbf{c}_m^\top (\rho_m \mathbf{R}(\theta_m))^\ell \mathbf{b}_m = \rho_m^\ell (A_m \cos(\ell\theta_m) + B_m \sin(\ell\theta_m)) \quad (16)$$

for scalar coefficients A_m, B_m determined by the write and read phases. A bank of such modes produces

$$h(\ell) = \sum_{m=1}^M \rho_m^\ell (A_m \cos(\ell\theta_m) + B_m \sin(\ell\theta_m)), \quad (17)$$

a complex-pole filter bank. It is BIBO stable when every radius is strictly below one; unit-radius modes are non-expansive without input but accumulate persistent writes. This is Fourier-like, but it is not generally a Fourier transform of the entire history: damping makes it an online rational filter, and token-dependent angles make it nonstationary. Complex phase with decay is already an established sequence-model construction [2]; negative real eigenvalues are also known to improve linear-RNN state tracking [1]. The question here is specifically how these modes interact with Kalman associative writes, uncertainty, and content-addressed retrieval.

The distinction between frequency and timescale is important. A low-frequency aligned mode has $\cos \theta \approx 1$ and can be maximally visible at the next token. A high-frequency mode can alternate or cancel, but need not be short lived. A desired recent-versus-delayed split must be designed jointly through radius, frequency, and relative write/read phase.

4 Hiding in one quadrature and revealing in the other

Let $\mathbf{e}_{\text{vis}} = (1, 0)^\top$ and $\mathbf{e}_{\text{hid}} = (0, 1)^\top$. An aligned immediate path has

$$\mathbf{b} = \mathbf{c} = \mathbf{e}_{\text{vis}}, \quad h_{\text{direct}}(\ell) = \rho^\ell \cos(\ell\theta). \quad (18)$$

In particular, $\theta = 0$ recovers ordinary geometric decay. A cross-quadrature path has

$$\mathbf{b} = \mathbf{e}_{\text{hid}}, \quad \mathbf{c} = \mathbf{e}_{\text{vis}}, \quad h_{\text{delay}}(\ell) = -\rho^\ell \sin(\ell\theta). \quad (19)$$

It is exactly invisible at insertion and grows as rotation moves it toward the visible axis. For $0 < \rho < 1$ and $0 < \theta < \pi$, treating ℓ as continuous, the first-lobe peak in magnitude is

$$\ell^* = \frac{1}{\theta} \arctan\left(\frac{\theta}{-\log \rho}\right), \quad (20)$$

which approaches $\pi/(2\theta)$ as $\rho \rightarrow 1$. The actual token lag is the maximizing nearby integer, so the continuous expression can be inaccurate near the Nyquist frequency. A slowly rotating channel is useful as a delayed channel only if its radius is large enough for the write to survive until this peak.

This is hidden state relative to the layer readout: it is currently unseen, not erased. For a sequence of read vectors $\mathbf{g}_{t+j}$, define the finite-horizon retrieval-observability Gramian

$$\mathbf{W}_o(t, L) = \sum_{j=0}^L \Phi_{t+j:t}^\top \mathbf{g}_{t+j} \mathbf{g}_{t+j}^\top \Phi_{t+j:t}. \quad (21)$$

A direction $\mathbf{z}$ can be hidden now because $\mathbf{g}_t^\top \mathbf{z} = 0$, yet remain recoverable if $\mathbf{z}^\top \mathbf{W}_o(t, L) \mathbf{z} > 0$. For one oscillator read through $\mathbf{e}_{\text{vis}}$, the two-step observability matrix is

$$\begin{bmatrix} \mathbf{e}_{\text{vis}}^\top \\ \mathbf{e}_{\text{vis}}^\top \rho \mathbf{R}(\theta) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \rho \cos \theta & -\rho \sin \theta \end{bmatrix}. \quad (22)$$

It has full rank whenever $\rho \sin \theta \neq 0$. Thus the latent quadrature is recoverable, although very small θ makes recovery poorly conditioned. The same rotation also causes periodic echoes; damping or a mixture of frequencies is needed when one delayed peak, rather than repeated resurfacing, is desired. This Gramian concerns the output read $\mathbf{g}$; detectability of the Kalman covariance is instead governed by the measurement feature used in the update.

5 A Fourier bank gives modulo-age selection

The frequency analogy can be made exact for age classes by spending state dimension on a temporal code. This section is an illustrative augmented-state, broadcast-gain mean recurrence, not yet the exact Kalman update derived in the next section: it changes the state dimension, write map, and lag-conditioned read in addition to changing $\mathbf{D}_t$. Let $\mathbf{Z}_{m,t} \in \mathbb{C}^{d_a \times d_v}$ be mode m , choose $\omega_m = 2\pi m/M$, and broadcast the same associative innovation $\mathbf{U}_t = \bar{\kappa}_t \delta_t^\top$ to every mode:

$$\mathbf{Z}_{m,t} = \rho e^{i\omega_m} \mathbf{Z}_{m,t-1} + \mathbf{U}_t, \quad m = 0, \dots, M-1. \quad (23)$$

Given a content query $\bar{\mathbf{q}}_t$ and a requested lag ℓ , read with the conjugate phase

$$\mathbf{o}_t(\ell) = \text{Re} \left[\frac{1}{M} \sum_{m=0}^{M-1} e^{-i\omega_m \ell} \mathbf{Z}_{m,t}^\top \bar{\mathbf{q}}_t \right]. \quad (24)$$

For integer age $n = t - \tau$ and requested integer lag ℓ , one write contributes the phase factor

$$\frac{1}{M} \sum_{m=0}^{M-1} e^{i\omega_m(n-\ell)} = \begin{cases} 1, & n \equiv \ell \pmod{M}, \\ 0, & \text{otherwise.} \end{cases} \quad (25)$$

Consequently,

$$\mathbf{o}_{t \leftarrow \tau}(\ell) = \rho^n (\bar{\boldsymbol{\kappa}}_\tau^\top \bar{\mathbf{q}}_t) \boldsymbol{\delta}_\tau \quad (26)$$

when $n \equiv \ell \pmod{M}$, and it cancels at the other represented age classes. Unrolling all writes makes the aliasing explicit:

$$\mathbf{o}_t(\ell) = \sum_{\substack{n \geq 0 \\ n \equiv \ell \pmod{M}}} \rho^n \mathbf{U}_{t-n}^\top \bar{\mathbf{q}}_t. \quad (27)$$

This is a modulo- M age-addressed accumulator, not an overwriting circular buffer. It selects one unique age only while the represented history is shorter than M ; afterward it sums all aliases, geometrically weighted when $\rho < 1$. Content similarity chooses *which* association to retrieve, while phase matching chooses *which age class* to expose. Complex arithmetic is not required: conjugate mode pairs correspond to the real 2×2 rotation blocks in (12); the DC mode, and the Nyquist mode for even M , are one-dimensional real exceptions.

A true length- M sliding Fourier delay line would have to remove the expired write,

$$\mathbf{Z}_{m,t} = \rho e^{i\omega_m} \mathbf{Z}_{m,t-1} + \mathbf{U}_t - \rho^M \mathbf{U}_{t-M}. \quad (28)$$

The expired write can be supplied by an external delay buffer, but it is also recoverable from the complete modal state. If $\mathbf{Z}_{m,t-1} = \sum_{n=0}^{M-1} \rho^n e^{i\omega_m n} \mathbf{U}_{t-1-n}$, then

$$\mathbf{U}_{t-M} = \rho^{-(M-1)} \frac{1}{M} \sum_{j=0}^{M-1} e^{i\omega_j} \mathbf{Z}_{j,t-1}. \quad (29)$$

Substitution into (28) gives the state-only update

$$\mathbf{Z}_{m,t} = \rho e^{i\omega_m} \mathbf{Z}_{m,t-1} + \mathbf{U}_t - \frac{\rho}{M} \sum_{j=0}^{M-1} e^{i\omega_j} \mathbf{Z}_{j,t-1}. \quad (30)$$

Thus exact overwrite remains fixed dimensional, but it is no longer an independent block-diagonal oscillator recurrence: the removal is a rank-one coupling across all modes. That coupling changes both the scan and the covariance structure developed for (13).

This construction answers the existence question, but it exposes the cost. It replicates a content address across M temporal modes, aliases ages separated by M , and requires the read controller to supply a lag or compatible phase. Unequal radii prevent the radial factor from leaving the Fourier sum and turn the exact age-class response into a soft kernel that depends on both age and requested lag. Learned complex weights can instead produce a general query-dependent response

$$h_t(n) = \text{Re} \left[\sum_m w_{t,m} \rho_m^n e^{in\omega_m} \right], \quad (31)$$

trading exact age-class selection for a lower-dimensional, task-adaptive temporal filter.

6 The corresponding Kalman model

A spectral state may use more rows than the original content key. It is therefore useful to separate the measurement address from the external key and the read address from the external query. Let

$$\mathbf{S}_t \in \mathbb{R}^{d_s \times d_v}, \quad \mathbf{P}_t \in \mathbb{R}^{d_s \times d_s}, \quad \mathbf{E}_t, \mathbf{F}_t \in \mathbb{R}^{d_s \times d_k}, \quad (32)$$

and define

$$\mathbf{h}_t = \mathbf{E}_t \mathbf{k}_t \in \mathbb{R}^{d_s}, \quad \mathbf{g}_t = \mathbf{F}_t \mathbf{q}_t \in \mathbb{R}^{d_s}, \quad (33)$$

where $\mathbf{E}_t$ chooses the measured modes and phases and $\mathbf{F}_t$ chooses the modes and phases visible to the query. Under a block-isotropic covariance the measurement lift also selects the corresponding write modes; under a dense covariance, cross-correlations can rotate the Kalman gain elsewhere. The generalized associative-memory filter is

$$\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}, \quad \hat{\mathbf{P}}_t = \mathbf{D}_t \mathbf{P}_{t-1} \mathbf{D}_t^\top + \boldsymbol{\Omega}_t, \quad (34)$$

$$\boldsymbol{\kappa}_t = \frac{\hat{\mathbf{P}}_t \mathbf{h}_t}{r_t + \mathbf{h}_t^\top \hat{\mathbf{P}}_t \mathbf{h}_t}, \quad \mathbf{S}_t = \hat{\mathbf{S}}_t + \boldsymbol{\kappa}_t (\mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{h}_t)^\top, \quad (35)$$

$$\mathbf{P}_t = (\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{h}_t^\top) \hat{\mathbf{P}}_t, \quad \mathbf{o}_t = \mathbf{S}_t^\top \mathbf{g}_t. \quad (36)$$

This is not a new filtering rule. It is the same Kalman filter with a richer latent geometry. The original model is recovered by $d_s = d_k$, $\mathbf{E}_t = \mathbf{F}_t = \mathbf{I}$, and a diagonal $\mathbf{D}_t$. The explicit $\mathbf{E}_t, \mathbf{F}_t$ make one limitation precise: $\mathbf{D}_t$ can preserve and move information, but it cannot by itself define what “hidden from this query” means.

The broadcast write in the Fourier example is not automatically the gain produced here. It is Kalman-consistent only under a symmetric measurement lift and covariance that yield identical gain blocks, or it can be used as a fixed-gain approximation. Its role above is to prove an age-selection property of an augmented spectral state, not to claim that $\mathbf{D}_t$ alone supplies a delay memory.

Rotational dynamics have a particularly convenient covariance approximation. For mode m , let its dimension be $d_m = 2r_m$. Assume the full covariance and process noise are block diagonal and isotropic inside every block,

$$\mathbf{P}_{t-1} = \text{blockdiag}_m(p_{t-1,m} \mathbf{I}_{d_m}), \quad \boldsymbol{\Omega}_t = \text{blockdiag}_m(\nu_{t,m} \mathbf{I}_{d_m}). \quad (37)$$

Then the prediction is scalar per mode:

$$\hat{\mathbf{P}}_t^{(m)} = (\rho_{t,m}^2 p_{t-1,m} + \nu_{t,m}) \mathbf{I}_{d_m}, \quad \hat{p}_{t,m} = \rho_{t,m}^2 p_{t-1,m} + \nu_{t,m}. \quad (38)$$

The rotation cancels from covariance prediction while still rotating the memory mean. The resulting gain is

$$\boldsymbol{\kappa}_t^{(m)} = \frac{\hat{p}_{t,m} \mathbf{h}_t^{(m)}}{r_t + \sum_j \hat{p}_{t,j} \|\mathbf{h}_t^{(j)}\|_2^2}. \quad (39)$$

An exact measurement update leaves the block-isotropic family: it creates rank-one anisotropy inside every observed block as well as correlations across blocks. This is the same closure problem that motivates projecting the current diagonal-information KDN rather than calling a diagonal posterior exact. A trace-matching projection back to the block-isotropic family would give

$$p_{t,m} = \hat{p}_{t,m} - \frac{\hat{p}_{t,m}^2 \|\mathbf{h}_t^{(m)}\|_2^2}{d_m \left(r_t + \sum_j \hat{p}_{t,j} \|\mathbf{h}_t^{(j)}\|_2^2 \right)}. \quad (40)$$

This projection is an approximation, not an exact block Kalman posterior. It is also a sequential coupled recurrence because its denominator depends on every block, so it does not give the independent Möbius scan used by the current diagonal KDN.

For a scan-compatible first model, use a calibrated scalar information state $c_{t,m} = 1/p_{t,m}$ per mode. Prediction is exact within the block-isotropic family,

$$\hat{c}_{t,m} = \frac{c_{t-1,m}}{\rho_{t,m}^2 + \nu_{t,m}c_{t-1,m}}, \quad \hat{p}_{t,m} = \frac{1}{\hat{c}_{t,m}}. \quad (41)$$

The measurement update is projected and standardized as

$$u_{t,m} = \frac{d_s}{d_m r_t} \|\mathbf{h}_t^{(m)}\|_2^2, \quad c_{t,m} = \hat{c}_{t,m} + u_{t,m}. \quad (42)$$

The literal trace projection of the information increment would omit the factor d_s . Equation (42) instead applies the same coordinate standardization as the current diagonal KDN: for an ℓ_2 -normalized feature spread across d_s coordinates, $\mathbb{E}[\|\mathbf{h}_t^{(m)}\|_2^2] \approx d_m/d_s$, so each block receives an expected $O(1/r_t)$ increment. This is a calibrated approximation, not an exact posterior projection.

Combining (41) and (42) gives the per-mode Möbius recurrence

$$c_{t,m} = \frac{(1 + u_{t,m}\nu_{t,m})c_{t-1,m} + u_{t,m}\rho_{t,m}^2}{\nu_{t,m}c_{t-1,m} + \rho_{t,m}^2}, \quad (43)$$

represented by the associative matrix

$$\mathbf{M}_{t,m} = \begin{bmatrix} 1 + u_{t,m}\nu_{t,m} & u_{t,m}\rho_{t,m}^2 \\ \nu_{t,m} & \rho_{t,m}^2 \end{bmatrix}. \quad (44)$$

The gains in (39) are then available from an independent prefix scan for each mode before the mean-memory scan. This is the uncertainty approximation assumed by the recommended spectral KDN below.

7 Scheduled resurfacing versus recall on cue

A fixed oscillator makes a memory visible according to elapsed time. That is useful for delay-sensitive prediction, but it does not by itself retrieve an arbitrary fact when a semantic cue arrives at an arbitrary future time. There are three increasingly expressive ways to add cue control.

Query-controlled observability. Let the query projection emit both quadratures and a mode gate. At ordinary tokens it can suppress persistent modes; at a recall cue it can read the matching hidden phase. In (36), this is a token-dependent $\mathbf{F}_t$ and requires no state-dependent transition. It is the simplest and most scan-friendly mechanism. The usual learned query projection can already implement part of this behavior, but an explicit per-mode gate makes the intended separation testable.

Input-conditioned phase increments. Use

$$\rho_{t,m} = \exp(-\Delta_{t,m}\lambda_m), \quad \theta_{t,m} = \Delta_{t,m}\omega_m + \gamma_{t,m}, \quad (45)$$

where $\Delta_{t,m} > 0$ is an input-dependent step size, $\lambda_m \geq 0$ is a decay rate, ω_m is a base frequency, and $\gamma_{t,m}$ is a bounded input-dependent phase kick. A retrieval cue can rotate a latent mode toward the visible phase before the current read. Because the coefficients depend only on input-known quantities, log-radii and angles still compose by prefix addition. The caveat is that one phase kick rotates every association in that mode; content selection must still come from the key/query match.

Active/archive routing. Allocate paired active and archive subspaces. A write gate routes durable information toward the archive quadrature, and a read gate chooses when the archive participates in the output. Continuous $\rho \mathbf{R}(\theta) \otimes \mathbf{I}$ gives scheduled, periodic exchange between active and archive coordinates; it does not keep an item archived until a cue. Cue-controlled archive semantics require either $\theta_t = 0$ between explicit transfer events or a query gate that fully suppresses the archive until retrieval. The lifts $\mathbf{E}_t$ and $\mathbf{F}_t$ then provide content-conditioned storage and recall. This gives the clearest semantics for “hide and later query,” at the cost of either a larger state or fewer content dimensions per subspace under a fixed state budget.

For arbitrary cue-driven recall, query-controlled observability should be present before dynamic phase kicks are added. Otherwise the model may learn a clock that solves fixed-delay training examples but fails when query times and recall order are randomized.

8 Parallelism and implementation constraints

The proposed transition remains structured, but it is not a drop-in use of the current KDA/GDN-2 delta-family kernels.

1. Applying block rotations to $\mathbf{S}_t$ costs $O(d_s d_v)$, the same order as diagonal decay, with a larger constant for paired multiply-adds. A dense learned change of basis would instead cost $O(d_s^2 d_v)$ and should be absorbed into projections or avoided.
2. Fixed-plane transitions compose associatively. Each mode can represent its prefix product by cumulative log-radius and cumulative phase, or by complex multiplication. This keeps the process prediction scan-friendly.
3. Once gains are input known, the full mean recurrence is the affine scan

$$\mathbf{S}_t = (\mathbf{I} - \kappa_t \mathbf{h}_t^\top) \mathbf{D}_t \mathbf{S}_{t-1} + \kappa_t \mathbf{v}_t^\top. \quad (46)$$

It is associative in principle, but efficient DeltaNet chunking exploits more structure than abstract associativity. The paired/complex key scores and decay normalization require a new derivation and kernel; equal asymptotic work does not imply equal throughput.

4. A block-isotropic uncertainty prediction stays scalar per mode, as shown in (38), and the calibrated update (43) supplies a per-mode gain scan. A full 2×2 covariance is a useful next step for a scalar-content plane. When $r_m > 1$, an exact update additionally creates correlations among content directions; cross-mode correlations still require projection or dense filtering.
5. If $\theta_{t,m}$ depends on the sequential memory $\mathbf{S}_{t-1}$, it is not known before the memory scan. Initial models should use fixed angles, token-local angles, or angles derived from separately scan-computable prefix statistics.

A conservative development ladder is

Stage	Transition	Question isolated
A0	pair-tied channels, $\theta = 0$	matched no-rotation control
A1	fixed θ_m , learned ρ_m	do complex poles help?
A2	fixed poles, learned write/read phase	can state be hidden then read?
A3	query-dependent read phase	can recall occur on an arbitrary cue?
A4	token-dependent phase increments	does dynamic reorganization help?

9 Limits and failure modes

1. **No extra capacity for free.** A fixed-dimensional linear state cannot retain an unbounded number of independent values losslessly. Oscillation changes temporal coding and interference; it does not eliminate superposition capacity limits.
2. **Aliasing.** A finite frequency grid confuses ages modulo its period. Damping, multiple incommensurate frequencies, resets, or a sufficiently large temporal code can soften this, but each choice has a state or optimization cost.
3. **Persistent noise.** With $\rho = 1$, additive writes and process noise do not decay. In a unitary mode that remains unobserved, covariance can grow without bound. Stable operation needs $\rho < 1$, reliable overwrite/reset behavior, or sufficiently frequent observability.
4. **Phase mismatch.** Rotating the memory address without a compatible key/query lift can make an association inaccessible rather than usefully latent. The transition, observation feature, and read feature must be designed together.
5. **Periodic leakage.** One oscillator repeatedly reveals a stored value. A mixture can localize one lag through interference, but imperfect learned phases may produce periodic false recalls.
6. **Optimization collapse.** Learned angles may collapse to zero, reducing the model to ordinary decay, or cluster at a few phases. Fixed frequency initialization and explicit diagnostics are needed before concluding that the spectral model is unused.

10 Falsifiable experiments

All comparisons should match total recurrent state dimension, parameter count, and, where possible, FLOPs. An augmented Fourier bank is otherwise advantaged simply by storing more numbers.

Lag-resolved associative recall. Train MQAR with a broad distribution of write–query gaps and report accuracy by exact lag, including lags outside the training range. Compare ordinary untied diagonal KDN/KDA, pair-tied $\theta = 0$ modes with exactly the same lifts and query gates as the spectral model, signed-real $\theta \in \{0, \pi\}$ modes, fixed oscillators, learned write/read phases, and query-demodulated modes. Periodic accuracy dips would directly reveal aliasing or phase mismatch.

Survival versus visibility probe. Write one tagged association, feed controlled distractors, and record both the process-only lineage $\Phi_{t:\tau}\kappa_\tau$ and the actual closed-loop lineage $\Psi_{t:\tau}\kappa_\tau$. For each, measure norm and visibility under a fixed read phase; an intervention difference between runs with and without the tagged write is an equivalent empirical measurement. Holding the read fixed prevents changing query routing from creating an apparent delayed peak even for diagonal decay. The defining spectral hypothesis is that closed-loop lineage norm can remain high while current visibility is low, followed by a later rise in visibility.

Cue-controlled recall. Randomize query times and the order in which stored items are requested. This prevents a fixed oscillator clock from solving the task. Ablate the temporal read phase, content router, and input-conditioned phase kick separately. A useful archive must respond to the cue rather than only to elapsed time.

Collision and capacity tests. At fixed lag, vary the number of simultaneously stored key–value pairs and the number of distractors. This tests whether improved lag selectivity is purchased by a loss of content capacity. Measure effective state rank and per-mode write/read energy in addition to task accuracy.

Core ablations and diagnostics. The essential controls are $\theta = 0$ damping only, signed-real $\theta \in \{0, \pi\}$, $\rho = 1$ rotation only, damped rotation, aligned versus cross-quadrature routing, fixed versus learned versus token-dependent angles, and diagonal versus block-isotropic uncertainty. Log learned poles (ρ_m, θ_m) , half-lives, phase usage, process-only and closed-loop impulse responses, state norms, covariance scale, throughput, and gradient stability.

11 Recommended first model

The first implementation should be a *spectral KDN* rather than a fully general learned transition:

1. pair fixed state planes and use $\mathbf{D}_t^{(m)} = \rho_{t,m} \mathbf{R}(\theta_m) \otimes \mathbf{I}_{r_m}$;
2. retain zero-angle modes spanning both fast and slow radii, and include the pair-tied zero-angle model with identical lifts and gates as the main no-rotation control;
3. initialize the remaining angles from log-spaced target delays L_m through $\theta_m \approx \pi/(2L_m)$ and give them radii whose lifetimes exceed L_m ;
4. let key and query projections learn the two quadratures, with an explicit query gate over modes;
5. use block-isotropic process noise and the calibrated Möbius information scan in (43), so the gain is available before the mean scan; and
6. test variable-lag and randomized-cue recall before introducing token-dependent phase kicks.

The central claim is deliberately narrow. Positive geometric decay decides how much of a memory remains. Damped rotations add a separate mechanism that decides where the surviving memory sits relative to the current readout. With phase-aware write and read maps, this can hide an association in a latent quadrature and later expose it. A complete Fourier bank gives exact selection of an age class, but without explicit overwrite it accumulates aliases rather than becoming a finite delay buffer. Whether the spectral process helps language modeling depends on closed-loop survival, content routing, cue-controlled observability, interference, and efficient kernels, all of which the proposed experiments measure directly.

References

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